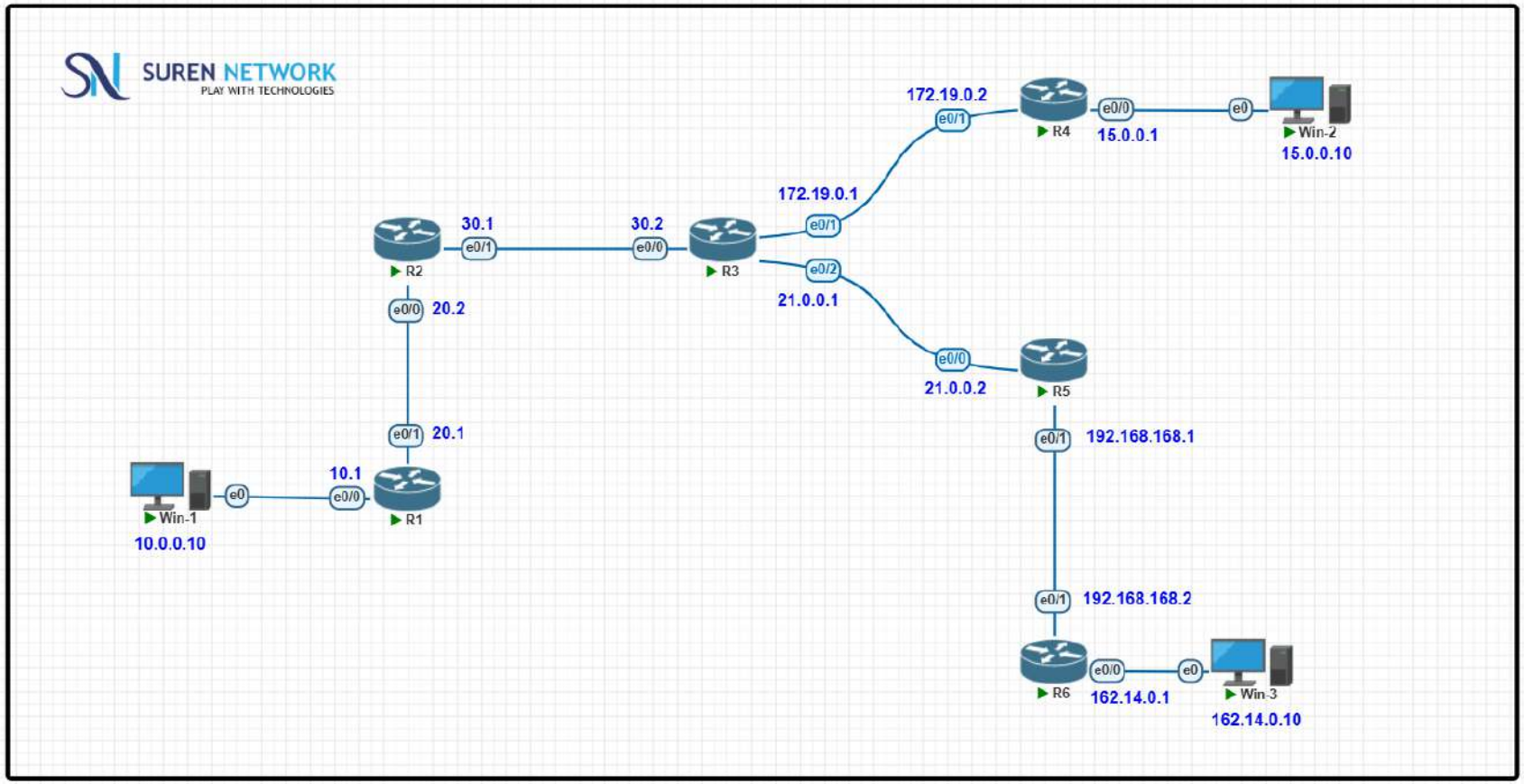
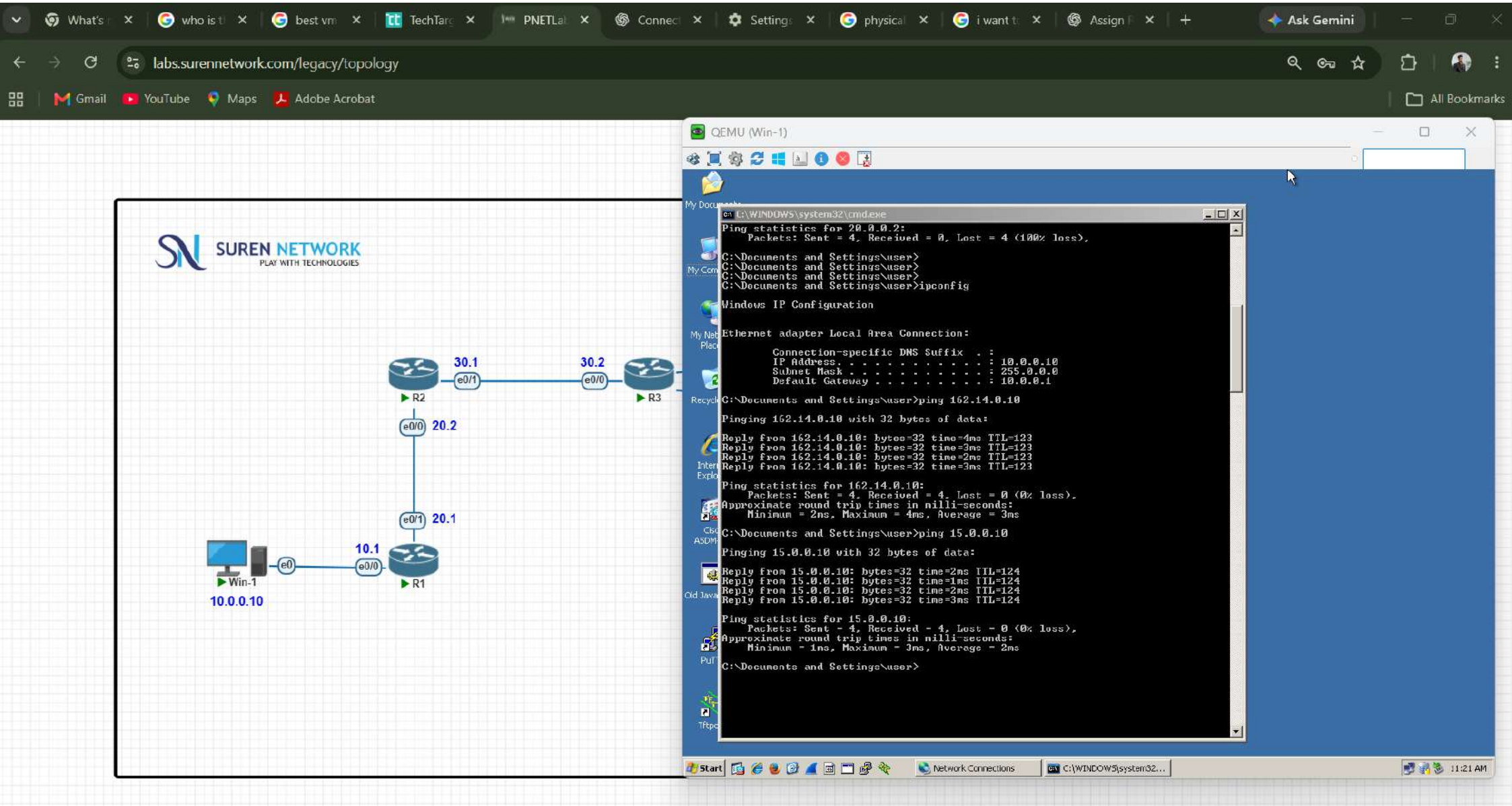

SUREN NETWORK
PLAY WITH TECHNOLOGIES
172.19.0.2
e0/1
R4
e0/0
15.0.0.1
e0
Win-2
15.0.0.10
172.19.0.1
e0/1
30.1
e0/1
R2
30.2
e0/0
R3
e0/2
21.0.0.1
e0/0
20.2
21.0.0.2
R5
e0/1
20.1
e0/1
192.168.168.1
10.1
e0/0
e0
Win-1
10.0.0.10
R1
e0/1
192.168.168.2
R6
e0/0
162.14.0.1
e0
Win-3
162.14.0.10


labs.surennetwork.com/legacy/topology
SUREN NETWORK
PLAY WITH TECHNOLOGIES
R2
R3
R1
Win-1
10.0.0.10
30.1
e0/1
30.2
e0/0
20.2
20.1
10.1
e0
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Ping statistics for 20.0.0.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IP Address. . . . . : 10.0.0.10
Subnet Mask . . . . . : 255.0.0.0
Default Gateway . . . . . : 10.0.0.1
C:\Documents and Settings\user>ping 162.14.0.10
Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Ping statistics for 162.14.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Ping statistics for 15.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>
Start
Network Connections
11:21 AM

R3

```
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 30.0.0.1
S     15.0.0.0/8 [1/0] via 172.19.0.2
S     20.0.0.0/8 [1/0] via 30.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
S     162.14.0.0/16 [1/0] via 21.0.0.2
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.1/32 is directly connected, Ethernet0/1
S     192.168.168.0/24 [1/0] via 21.0.0.2
R3#
R3#
```

R1

```
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
S     15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
S     21.0.0.0/8 [1/0] via 20.0.0.2
S     30.0.0.0/8 [1/0] via 20.0.0.2
S     162.14.0.0/16 [1/0] via 20.0.0.2
S     172.19.0.0/16 [1/0] via 20.0.0.2
S     192.168.168.0/24 [1/0] via 20.0.0.2
R1#
```

R2

```
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 20.0.0.1
S     15.0.0.0/8 [1/0] via 30.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
S     21.0.0.0/8 [1/0] via 30.0.0.2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
S     162.14.0.0/16 [1/0] via 30.0.0.2
S     172.19.0.0/16 [1/0] via 30.0.0.2
S     192.168.168.0/24 [1/0] via 30.0.0.2
R2#
```

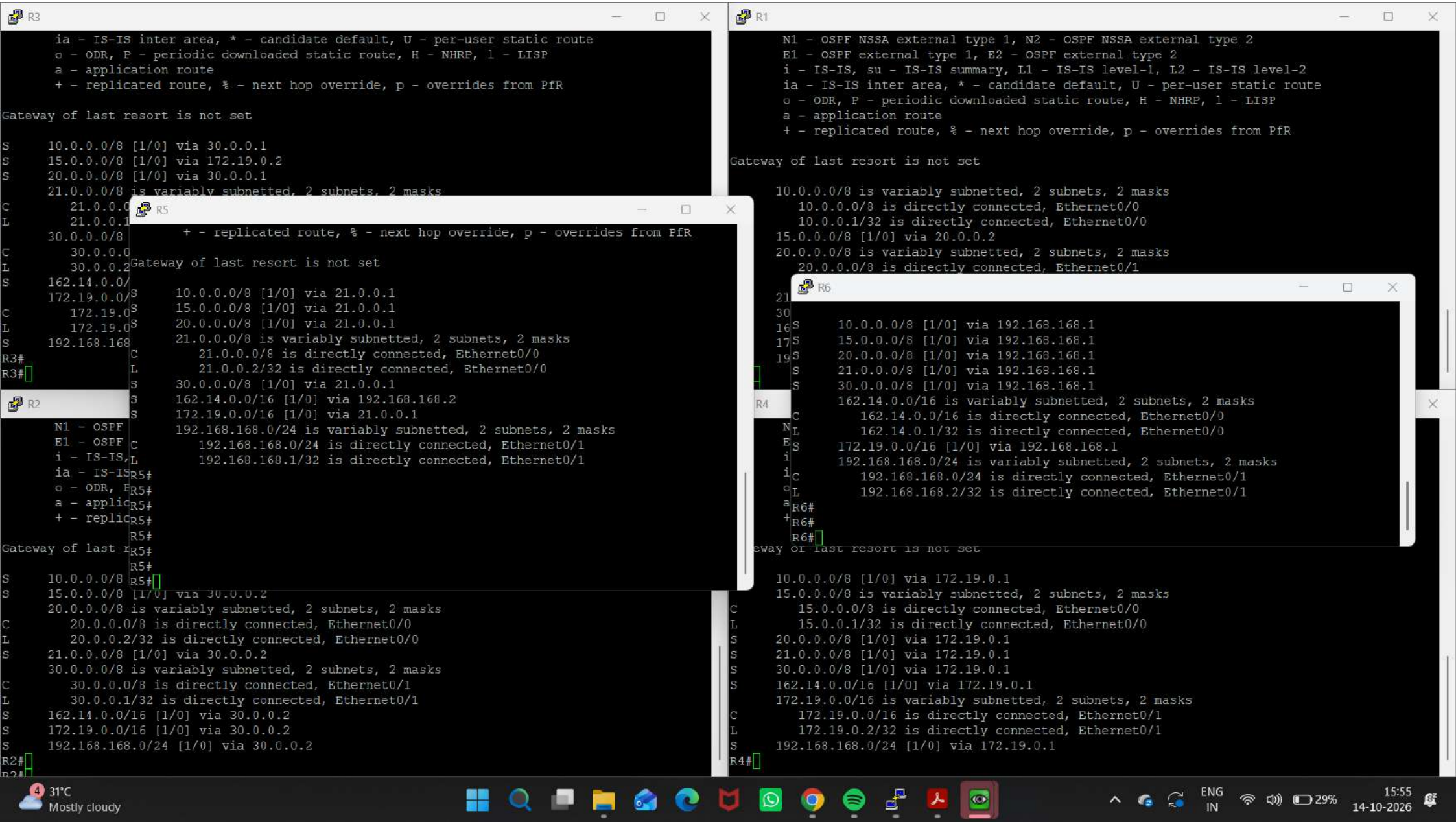
R4

```
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 172.19.0.1
      15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        15.0.0.0/8 is directly connected, Ethernet0/0
L        15.0.0.1/32 is directly connected, Ethernet0/0
S     20.0.0.0/8 [1/0] via 172.19.0.1
S     21.0.0.0/8 [1/0] via 172.19.0.1
S     30.0.0.0/8 [1/0] via 172.19.0.1
S     162.14.0.0/16 [1/0] via 172.19.0.1
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.2/32 is directly connected, Ethernet0/1
S     192.168.168.0/24 [1/0] via 172.19.0.1
R4#
```

31°C
Mostly cloudy
ENG
IN
29%
15:54
14-10-2026


R3
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0
L 21.0.0.1
30.0.0.0/8
C 30.0.0.0
L 30.0.0.2
S 162.14.0.0/
172.19.0.0/
C 172.19.0
L 172.19.0
S 192.168.168
R3#
R3#
R5
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/0
L 21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.19.0.0/16 [1/0] via 21.0.0.1
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#
R5#
R5#
R5#
R5#
R5#
R5#
R2
N1 - OSPF
E1 - OSPF
i - IS-IS,
ia - IS-IS
o - ODR, P
a - applic
+ - replic
Gateway of last r
S 10.0.0.0/8
S 15.0.0.0/8 [1/0] via 30.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2
R2#
R1
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
10.0.0.0/8 is directly connected, Ethernet0/0
10.0.0.1/32 is directly connected, Ethernet0/0
15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
20.0.0.0/8 is directly connected, Ethernet0/1
R6
S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
R6#
R4
eway of last resort is not set
10.0.0.0/8 [1/0] via 172.19.0.1
15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/8 is directly connected, Ethernet0/0
L 15.0.0.1/32 is directly connected, Ethernet0/0
S 20.0.0.0/8 [1/0] via 172.19.0.1
S 21.0.0.0/8 [1/0] via 172.19.0.1
S 30.0.0.0/8 [1/0] via 172.19.0.1
S 162.14.0.0/16 [1/0] via 172.19.0.1
172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.2/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 172.19.0.1
R4#
31°C
Mostly cloudy
ENG
IN
29%
15:55
14-10-2026